

The empirical recurrence for OEIS A237399, proved by transfer matrix

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Abstract

OEIS A237399 counts arrays over $\{0, \dots, 3\}$ subject to a condition on the order statistics of every 2×2 subblock, and carries an empirical recurrence of order 40 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is local to two consecutive rows, so the count is a walk count on $S = 256$ states and is C -finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A237399 is “Number of $(n+1) \times (1+1)$ 0..3 arrays with the maximum plus the upper median plus the lower median minus the minimum of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

256, 1540, 9296, 58088, 363220, 2273712, 14242608, 89287168, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 1944*a(n-4) - 669984*a(n-8) + 82555584*a(n-12) - 4702054656*a(n-16) + 139358364672*a(n-20) - 2211858219008*a(n-24) + 17888510001152*a(n-28) - 60723254525952*a(n-32) + 30235468759040*a(n-36) - 3292511862784*a(n-40)$ for $n > 43$.

As of the “Last modified” line on the live entry (Feb 07 2014 19:23:52, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The 2×2 subblocks of an $(n+1) \times 2$ array form an $n \times 1$ grid, and each carries the value $\sigma = -a + b + c + d$, where $a \leq b \leq c \leq d$ are its four entries sorted — its minimum, lower median, upper median and maximum. The entry asks that the values of subblocks that are neighbours — horizontally or vertically — differ by exactly 1. That is sharper than merely differing, and it is a condition on a pair of adjacent subblocks, so it is local to the grid.

A single subblock has no neighbour of either kind, so at $n = 1$ the condition is vacuous and $a(1)$ must be the number of 2×2 arrays over $\{0, \dots, 3\}$, namely $4^4 = 256$. The entry’s first term is 256. That is the cheapest check that the subblock grid has been read with the right shape, and it is the first thing the model was made to reproduce.

3 The count is a walk count

A subblock's value is decided by two consecutive rows of the array, and the vertical condition compares two consecutive rows of the subblock grid, so the state is the PAIR of consecutive array rows. A step appends a row: that fixes a new subblock row, whose own neighbouring values must differ by exactly 1 along it, and which must differ by exactly 1 from the previous subblock row in every position. An $(n + 1)$ -row array is a walk of $n - 1$ steps, so

$$a(n) = \iota^\top M^{n-1} \tau, \quad \iota = \tau = (1, \dots, 1)^\top,$$

over the $S = 256$ pairs whose own subblock row already satisfies the condition along itself. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 1. *Let $q(t) = t^{40} - \sum_i c_i t^{40-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+40} - \sum_i c_i M^{j+40-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 1944 a(n-4) - 669984 a(n-8) + 82555584 a(n-12) - 4702054656 a(n-16) + 139358364672 a(n-20) - 2211$$

for every $n > 43$.

Proof. The vector $w = q(M) \tau$ was formed by 40 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 43 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
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